PRACTICAL REPORT

ON

PPSCSMAJM402: BIG DATA ANALYTICS

SUBMITTED BY

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ROLL NO: 18

SUBMITTED TO

MS. RASIKA SAWANT

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CONDUCTED AT
CHIKITSAK SAMUHA'S

S. S. & L.S. PATKAR COLLEGE OF ARTS & SCIENCE

AND

V. P. VARDE COLLEGE OF COMMERCE & ECONOMICS

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CERTIFICATE

Certified that such of the experiments as have been duly signed

were performed by Mr./Miss _____

Roll No. _____ *of* _____ *class* _____

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Practical No. 1

Aim : Installing and setting environment variable for working with apache Hadoop

Step 1: Open Window Powershell and run as admin and then type

```
PS C:\Windows\system32> wsl --install
Ubuntu is already installed.
Launching Ubuntu...
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Feb 15 12:10:48 UTC 2026

  System load:  1.75                Processes:             55
  Usage of /:   2.8% of 1006.85GB   Users logged in:       0
  Memory usage: 7%                  IPv4 address for eth0: 172.23.24.114
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
```

Step 2: Type Command - sudo apt-get update

```
siddhesh@DESKTOP-74T5BGT:~$ sudo apt-get update
[sudo] password for siddhesh:
Get:1 https://deb.nodesource.com/node_16.x nodistro InRelease [12.1 kB]
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu jammy InRelease [18.3 kB]
Get:6 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu jammy/main amd64 Packages [2580 B]
Get:7 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2959 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3226 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [492 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.1 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [5260 kB]
```

Step 3: Type Command : sudo apt-get install openjdk-8-jdk

```
siddhesh@DESKTOP-74T5BGT:~$ sudo apt-get install openjdk-8-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java fonts-dejavu-extra java-common libasound2 libasound2-data
  libasyncns0 libatk-wrapper-java libatk-wrapper-java-jni libflac8 libfontenc1 libgail-common libgail18 libgif7
  libgtk2.0-0 libgtk2.0-bin libgtk2.0-common libice-dev libice6 libnspr4 libnss3 libogg0 libopus0 libpcsclite1
  libpthread-stubs0-dev libpulse0 libsm-dev libsm6 libsndfile1 libvorbis0a libvorbisenc2 libx11-dev libxau-dev libxaw
  libxcb-shape0 libxcb1-dev libxdmcp-dev libxft2 libxkbfile1 libxmu6 libxt-dev libxt6 libxv1 libxxf86dga1
  openjdk-8-jdk-headless openjdk-8-jre openjdk-8-jre-headless x11-utils x11proto-dev xorg-sgml-doctools xtrans-dev
Suggested packages:
  default-jre libasound2-plugins alsa-utils gvfs libice-doc opus-tools pcscd pulseaudio libsm-doc libx11-doc
  libxcb-doc libxt-doc openjdk-8-demo openjdk-8-source visualvm libnss-mdns fonts-nanum fonts-ipafont-gothic
  fonts-ipafont-mincho fonts-wqy-microhei fonts-wqy-zenhei fonts-indic mesa-utils
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java fonts-dejavu-extra java-common libasound2 libasound2-data
  libasyncns0 libatk-wrapper-java libatk-wrapper-java-jni libflac8 libfontenc1 libgail-common libgail18 libgif7
  libgtk2.0-0 libgtk2.0-bin libgtk2.0-common libice-dev libice6 libnspr4 libnss3 libogg0 libopus0 libpcsclite1
  libpthread-stubs0-dev libpulse0 libsm-dev libsm6 libsndfile1 libvorbis0a libvorbisenc2 libx11-dev libxau-dev libxaw
```

Step 4: Type Command : sudo addgroup hadoop

```
siddhesh@DESKTOP-74T5BGT:~$ sudo addgroup hadoop
Adding group `hadoop' (GID 1002) ...
Done.
siddhesh@DESKTOP-74T5BGT:~$
```

Step 5: Type Command :

sudo adduser -ingroup hadoop siddhesh

sudo usermod -aG sudo siddhesh

su siddhesh

```
siddhesh@DESKTOP-74T5BGT:~$ sudo adduser -ingroup hadoop siddhesh
adduser: The user `siddhesh' already exists.
siddhesh@DESKTOP-74T5BGT:~$ sudo usermod -aG sudo siddhesh
siddhesh@DESKTOP-74T5BGT:~$ su siddhesh
Password:
siddhesh@DESKTOP-74T5BGT:~$
```

Step 6: Type Command : ssh-keygen -t rsa -P ""

```
siddhesh@DESKTOP-74T5BGT:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/siddhesh/.ssh/id_rsa):
Created directory '/home/siddhesh/.ssh'.
Your identification has been saved in /home/siddhesh/.ssh/id_rsa
Your public key has been saved in /home/siddhesh/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:9GS/XLEk8w9x5NT3sJ2FlU5f1PGq1Vlon/FOu7vit1U siddhesh@DESKTOP-74T5BGT
The key's randomart image is:
+---[RSA 3072]----+
|               =X|
|              oBB|
|         . o o O=%|
|        . + . *.@X|
|         S . . BoE|
|            . = =o|
|             +  .+|
|              . .o|
|             ..o=+|
+----[SHA256]-----+
siddhesh@DESKTOP-74T5BGT:~$
```

Step 7: Type Command :

touch $HOME/.ssh/authorized_keys

cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys

```
+----[SHA256]-----+
siddhesh@DESKTOP-74T5BGT:~$ touch $HOME/.ssh/authorized_keys
siddhesh@DESKTOP-74T5BGT:~$ cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
siddhesh@DESKTOP-74T5BGT:~$
```

Step 8: Type Command : sudo nano /etc/sysctl.conf

Code to write inside the file

net.ipv6.conf.all.disable_ipv6=1
net.ipv6.conf.default.disable_ipv6=1
net.ipv6.conf.lo.disable_ipv6=1

```
siddhesh@DESKTOP-74T5BGT: ~
  GNU nano 6.2                                                    /etc/sysctl.conf
# settings are disabled so review and enable them as needed.
#
# Do not accept ICMP redirects (prevent MITM attacks)
#net.ipv4.conf.all.accept_redirects = 0
#net.ipv6.conf.all.accept_redirects = 0
# _or_
# Accept ICMP redirects only for gateways listed in our default
# gateway list (enabled by default)
# net.ipv4.conf.all.secure_redirects = 1
#
# Do not send ICMP redirects (we are not a router)
#net.ipv4.conf.all.send_redirects = 0
#
# Do not accept IP source route packets (we are not a router)
#net.ipv4.conf.all.accept_source_route = 0
#net.ipv6.conf.all.accept_source_route = 0
#
# Log Martian Packets
#net.ipv4.conf.all.log_martians = 1
#

###################################################################
# Magic system request Key
# 0=disable, 1=enable all, >1 bitmask of sysrq functions
# See https://www.kernel.org/doc/html/latest/admin-guide/sysrq.html
# for what other values do
#kernel.sysrq=438

net.ipv6.conf.all.disable_ipv6=1
net.ipv6.conf.default.disable_ipv6=1
net.ipv6.conf.lo.disable_ipv6=1
```

Step 9: Type Command :
cd /usr/local
sudo wget https://archive.apache.org/dist/hadoop/core/hadoop-3.3.1/hadoop-3.3.1.tar.gz

```
siddhesh@DESKTOP-74T5BGT:~$ cd /usr/local
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo wget https://archive.apache.org/dist/hadoop/core/hadoop-3.3.1/hadoop-3.3.1.tar.gz
--2026-02-15 17:57:03--  https://archive.apache.org/dist/hadoop/core/hadoop-3.3.1/hadoop-3.3.1.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response...
```

Step 10: Extract Hadoop Archive Type Command : sudo tar xzvf hadoop-3.3.1.tar.gz

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo wget https://archive.apache.org/dist/hadoop/core/hadoop-3.3.1/hadoop-3.3.1.tar.gz
--2026-02-15 18:42:28--  https://archive.apache.org/dist/hadoop/core/hadoop-3.3.1/hadoop-3.3.1.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 605187279 (577M) [application/x-gzip]
Saving to: 'hadoop-3.3.1.tar.gz.2'

hadoop-3.3.1.tar.gz.2                          100%[===================================================================================================>] 577.15M   951KB/s    in 8m 24s

2026-02-15 18:50:53 (1.15 MB/s) - 'hadoop-3.3.1.tar.gz.2' saved [605187279/605187279]

siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 11: Type Command :
sudo mv hadoop-3.3.1 hadoop
sudo chown -R siddhesh:hadoop hadoop

```
tar: Error is not recoverable: exiting now
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo mv hadoop-3.3.1 hadoop
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo chown -R siddhesh:hadoop hadoop
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 12: Type Command : nano ~/.bashrc
Set Hadoop and Java paths
export HADOOP_HOME=/usr/local/hadoop
export JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-amd64

Define Hadoop component paths
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native

Add Hadoop to system PATH
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
Set Hadoop JVM options
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"

```
tar: Error is not recoverable: exiting now
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo mv hadoop-3.3.1 hadoop
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo chown -R siddhesh:hadoop hadoop
siddhesh@DESKTOP-74T5BGT:/usr/local$ nano ~/.bashrc
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 13: Type Command : source ~/.bashrc

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ source ~/.bashrc
siddhesh@DESKTOP-74T5BGT:/usr/local$ cd /usr/local/hadoop/etc/hadoop
bash: cd: /usr/local/hadoop/etc/hadoop: No such file or directory
```

Step 14: Type Command :
cd /usr/local/hadoop/etc/hadoop
nano hadoop-env.sh

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ source ~/.bashrc
siddhesh@DESKTOP-74T5BGT:/usr/local$ cd /usr/local/hadoop/etc/hadoop
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ nano hadoop-env.sh
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$
```

Step 15: Type Command :
sudo mkdir -p /app/hadoop/tmp
sudo chown siddhesh:hadoop /app/hadoop/tmp

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ source ~/.bashrc
siddhesh@DESKTOP-74T5BGT:/usr/local$ cd /usr/local/hadoop/etc/hadoop
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ nano hadoop-env.sh
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ sudo mkdir -p /app/hadoop/tmp
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ sudo chown siddhesh:hadoop /app/hadoop/tmp
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$
```

Step 16: Type Command : nano core-site.xml

```
  GNU nano 6.2
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->


<configuration>
<property>
<name>hadoop.tmp.dir</name>
<value>/app/hadoop/tmp</value>
<description>A base for other temporary directories.</description>

</property>
<property>
<name>fs.default.name</name>
<value>hdfs://127.0.0.1:54310</value>
</property>

</configuration>
```

Step 17: Type Command : nano mapred-site.sh

siddhesh@DESKTOP-74T5BGT: /usr/local/hadoop/etc/hadoop

```
  GNU nano 6.2
<property>
<name>mapred.job.tracker</name>
<value>hdfs://127.0.0.1:54311</value>
</property>
```

Step 18: Type Command : nano hdfs-site.xml

```
  GNU nano 6.2
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>

</configuration>
```

Step 19: Type Command : sudo apt-get install ssh

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ nano hdfs-site.xml
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ sudo apt-get install ssh
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libwrap0 ncurses-term openssh-server openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  libwrap0 ncurses-term openssh-server openssh-sftp-server ssh ssh-import-id
0 upgraded, 6 newly installed, 0 to remove and 90 not upgraded.
Need to get 804 kB of archives.
After this operation, 6296 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Step 20: Type Command : sudo apt update && sudo apt install openssh-server -y

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ sudo apt update && sudo apt install openssh-server -y
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 https://deb.nodesource.com/node_16.x nodistro InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:6 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu jammy InRelease
```

Step 21: Type Command : ssh localhost

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/etc/hadoop$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:/xR26mtvDSDKeBGWWHpl0ekUr9gxAE32r6yJK/pbD38.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Step 22: Type Command : sudo service ssh restart

```
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Sun Feb 15 17:42:36 2026
siddhesh@DESKTOP-74T5BGT:~$ sudo service ssh restart
[sudo] password for siddhesh:
siddhesh@DESKTOP-74T5BGT:~$
```

Step 23: Type Command : cd /usr/local/hadoop/sbin

```
siddhesh@DESKTOP-74T5BGT:~$ sudo service ssh restart
[sudo] password for siddhesh:
siddhesh@DESKTOP-74T5BGT:~$ cd /usr/local/hadoop/sbin
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/sbin$
```

Step 24: Type Command : hdfs namenode -format

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/sbin$ hdfs namenode -format
WARNING: /usr/local/hadoop/logs does not exist. Creating.
2026-02-15 19:34:06,571 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = DESKTOP-74T5BGT/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.3.1
STARTUP_MSG:   classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/jsch-0.1.55.jar:/usr/local/h
mmon/lib/javax.servlet-api-3.1.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-core-2.10.5.jar:/usr/local/hadoop/share
ommons-net-3.6.jar:/usr/local/hadoop/share/hadoop/common/lib/token-provider-1.0.1.jar:/usr/local/hadoop/share/hadoop/common/li
1.19.jar:/usr/local/hadoop/share/hadoop/common/lib/j2objc-annotations-1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jcip-a
/usr/local/hadoop/share/hadoop/common/lib/paranamer-2.3.jar:/usr/local/hadoop/share/hadoop/common/lib/animal-sniffer-annotatio
hadoop/share/hadoop/common/lib/jetty-util-9.4.40.v20210413.jar:/usr/local/hadoop/share/hadoop/common/lib/stax2-api-4.2.1.jar:/
adoop/share/hadoop/common/lib/curator-recipes-4.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-client-1.0.1.jar:/usr/l
op/share/hadoop/common/lib/jaxb-impl-2.2.3-1.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-webapp-9.4.40.v20210413.jar:/
oop/common/lib/curator-framework-4.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-lang3-3.7.jar:/usr/local/hadoop/s
adoop/common/lib/commons-cli-1.2.jar:/usr/local/hadoop/share/hadoop/common/lib/dnsjava-2.1.7.jar:/usr/local/hadoop/share/hadoo
tivation-api-1.2.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jsp-api-2.1.jar:/usr/local/hadoop/share/hadoop/common/lib/com
```

Step 25: Type Command : start-all.sh

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hadoop/sbin$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as siddhesh in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [DESKTOP-74T5BGT]
DESKTOP-74T5BGT: Warning: Permanently added 'desktop-74t5bgt' (ED25519) to the list of known hosts.
2026-02-15 19:35:19,163 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Starting nodemanagers
```

Practical No. 2

Aim: Implementing Map-Reduce Program for Word Count problem

Step 1: Complete the hadoop env steps

su siddhesh

sudo service ssh restart

ssh localhost

start-all.sh

```
siddhesh@DESKTOP-74T5BGT:/$ su siddhesh
Password:
siddhesh@DESKTOP-74T5BGT:/$ sudo service ssh restart
siddhesh@DESKTOP-74T5BGT:/$ ssh localhost
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Feb 15 19:40:10 IST 2026

  System load:  0.06                Processes:             44
  Usage of /:   0.6% of 1006.85GB   Users logged in:       1
  Memory usage: 24%                 IPv4 address for eth0: 172.23.24.114
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Sun Feb 15 19:29:58 2026 from 127.0.0.1
siddhesh@DESKTOP-74T5BGT:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as siddhesh in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
```

Step 2: **Type Command**

mkdir Mapreduce

cd Mapreduce

```
Starting nodemanagers
localhost: nodemanager is running as process 5493.  Stop it first and ensure /tmp
siddhesh@DESKTOP-74T5BGT:~$ mkdir Mapreduce
siddhesh@DESKTOP-74T5BGT:~$ cd Mapreduce
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$
```

Step 3: **Create file :**

touch word_count_data.txt

nano word_count_data.txt

```
siddhesh@DESKTOP-74T5BGT: ~/Mapreduce
  GNU nano 6.2
Bus train
train car Bus
car Bus train bus
```

Step 4: **Try to view the data**

cat word_count_data.txt

```
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ nano word_count_data.txt
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ cat word_count_data.txt
Bus train
train car Bus
car Bus train bus
```

Step 5: **Create mapper python file**

touch mapper.py

nano mapper.py

```
import sys

for line in sys.stdin:
    line = line.strip()
    words = line.split()
    for word in words:
        print("%s\t%s" % (word, 1))
```

```
siddhesh@DESKTOP-74T5BGT: ~/Mapreduce
  GNU nano 6.2
import sys
for line in sys.stdin:
line = line.strip()
words = line.split()
for word in words:
print("%s\t%s" % (word, 1))
```

Step 6: **Try to run this code with txt file**

cat word_count_data.txt | python3 mapper.py

```
TabError: inconsistent use of tabs and spaces in indentation
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ nano mapper.py
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ cat word_count_data.txt | python3 mapper.py
Bus     1
train   1
train   1
car     1
Bus     1
car     1
Bus     1
train   1
bus     1
```

Step 7: **Create reducer python file**

touch reducer.py

nano reducer.py

```
import sys

word_counts = {}

for line in sys.stdin:
word, count = line.strip().split("\t")
word_counts[word] = word_counts.get(word, 0) + int(count)

for word, count in word_counts.items():
print(f"{word}\t{count}")
```

```
bus     1
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ touch reducer.py
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ nano reducer.py
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$
```

Step 8: **Try to run the txt file and both python code**

cat word_count_data.txt | python3 mapper.py | sort | python3 reducer.py

```
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ cat word_count_data.txt | python3 mapper.py | sort | python3 reducer.py
Bus     3
bus     1
car     2
train   3
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$
```

Practical No. 3

Aim: Implementing MapReduce program to sort text data.

Step 1: Complete the hadoop env steps

su hduser

sudo service ssh restart

ssh localhost

start-all.sh

```
siddhesh@DESKTOP-74T5BGT:/$ su siddhesh
Password:
siddhesh@DESKTOP-74T5BGT:/$ sudo service ssh restart
siddhesh@DESKTOP-74T5BGT:/$ ssh localhost
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Feb 15 19:40:10 IST 2026

  System load:  0.06                Processes:             44
  Usage of /:   0.6% of 1006.85GB   Users logged in:       1
  Memory usage: 24%                 IPv4 address for eth0: 172.23.24.114
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Sun Feb 15 19:29:58 2026 from 127.0.0.1
siddhesh@DESKTOP-74T5BGT:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as siddhesh in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
```

Step 2: **Make folder mapreduce and create a file in it named localfile.txt and open it**

mkdir mapreduce

cd mapreduce

touch localfile.txt

nano localfile.txt

```
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ mkdir mapreduce
siddhesh@DESKTOP-74T5BGT:~/Mapreduce$ cd mapreduce
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$ touch localfile.txt
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$ nano localfile.txt
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$
```

```
siddhesh@DESKTOP-74T5BGT: ~/Mapreduce/mapreduce
  GNU nano 6.2
banana
apple
cherry
watermelon
chickoo
mango
grapes
orange
gauva
kiwi
avacado
```

Step 3: **Try to view data in it**

cat localfile.txt

```
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$ cat localfile.txt
banana
apple
cherry
watermelon
chickoo
mango
grapes
orange
gauva
kiwi
avacado
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$
```

Step 4: **Create python file name mapper.py and open it**

touch mapper.py

nano mapper.py

siddhesh@DESKTOP-74T5BGT: ~/Mapreduce/mapreduce

```
  GNU nano 6.2
import sys

for line in sys.stdin:
    line = line.strip()
    print(line)
```

Step 5: **Open the file and run the python file**

cat localfile.txt | python3 mapper.py

```
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$ nano mapper.py
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$ cat localfile.txt | python3 mapper.py
banana
apple
cherry
watermelon
chickoo
mango
grapes
orange
gauva
kiwi
avacado
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$
```

Step 6: **Create python file name reducer.py and open it**

touch reducer.py

nano reducer.py

siddhesh@DESKTOP-74T5BGT: ~/Mapreduce/mapreduce

```
  GNU nano 6.2
import sys

sorted_lines = []

for line in sys.stdin:
    word = line.strip()
    sorted_lines.append(word)

sorted_lines.sort()

for line in sorted_lines:
    print(line)
```

Step 7: **Open the file and run the python file**

cat localfile.txt | python3 mapper.py | python3 reducer.py

```
siddhesh@DESKTOP-74T5BGT:~/Mapreduce/mapreduce$ cat localfile.txt | python3 mapper.py | python3 reducer.py
apple
avacado
banana
cherry
chickoo
gauva
grapes
kiwi
mango
orange
watermelon
```

Practical No. 4

Aim: Download and install Spark. Create Graphical data and access the graphical data using Spark.

Step 1: Open **Ubuntu (WSL)** terminal

siddhesh@DESKTOP-74T5BGT: ~

```
siddhesh@DESKTOP-74T5BGT:~$ su siddhesh
Password:
siddhesh@DESKTOP-74T5BGT:~$
```

Step 2: **Update packages**

sudo apt update && sudo apt upgrade -y

```
siddhesh@DESKTOP-74T5BGT:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for siddhesh:
Hit:1 https://deb.nodesource.com/node_16.x nodistro InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu jammy InRelease
Hit:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
```

Step 3: **Install Java (Required for Spark)** Spark works best with **Java 8 or 11.**

sudo apt install openjdk-11-jdk -y

```
Processing triggers for install-info (6.8-4build1) ...
siddhesh@DESKTOP-74T5BGT:~$ sudo apt install openjdk-11-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless
Suggested packages:
  openjdk-11-demo openjdk-11-source visualvm libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic
The following NEW packages will be installed:
  openjdk-11-jdk openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 119 MB of archives.
After this operation, 262 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 openjdk-11-jre-headless amd64 11.0.30+7-1ubuntu1~22.04 [42.6 MB]
```

Step 4: java -version

```
Processing triggers for hicolor-icon-theme (0.17-2) ...
siddhesh@DESKTOP-74T5BGT:~$ java -version
openjdk version "11.0.30" 2026-01-20
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu122.04, mixed mode, sharing)
siddhesh@DESKTOP-74T5BGT:~$
```

Step 5: **Install Python & pip**

sudo apt install python3 python3-pip -y

```
siddhesh@DESKTOP-74T5BGT:~$ sudo apt install python3 python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.10.6-1~22.04.1).
python3 set to manually installed.
The following additional packages will be installed:
  libexpat1-dev libjs-sphinxdoc libjs-underscore libpython3-dev libpython3.10-dev python3-dev python3-wheel python3.10-dev zlib1g-dev
The following NEW packages will be installed:
  libexpat1-dev libjs-sphinxdoc libjs-underscore libpython3-dev libpython3.10-dev python3-dev python3-pip python3-wheel python3.10-dev zlib1g-dev
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 7212 kB of archives.
After this operation, 31.0 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libexpat1-dev amd64 2.4.7-1ubuntu0.7 [148 kB]
```

Step 6: Verify :

python3 –version

pip3 –version

```
siddhesh@DESKTOP-74T5BGT:~$ python3 --version
Python 3.10.12
siddhesh@DESKTOP-74T5BGT:~$ pip3 --version
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)
siddhesh@DESKTOP-74T5BGT:~$
```

Step 7: Download and Install Apache Spark Download Spark

wget https://archive.apache.org/dist/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz

```
siddhesh@DESKTOP-74T5BGT:~$ wget https://archive.apache.org/dist/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz
--2026-02-15 20:18:35--  https://archive.apache.org/dist/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 400395283 (382M) [application/x-gzip]
Saving to: 'spark-3.5.0-bin-hadoop3.tgz'

spark-3.5.0-bin-hadoop3.tgz          100%[===================================================================================================>] 381.85M  3.41MB/s    in 3m 41s

2026-02-15 20:22:18 (1.73 MB/s) - 'spark-3.5.0-bin-hadoop3.tgz' saved [400395283/400395283]

siddhesh@DESKTOP-74T5BGT:~$
```

Step 8: **Extract Spark**

tar -xvzf spark-3.5.0-bin-hadoop3.tgz

```
spark-3.5.0-bin-hadoop3/bin/pyspark
spark-3.5.0-bin-hadoop3/bin/pyspark.cmd
spark-3.5.0-bin-hadoop3/bin/load-spark-env.sh
spark-3.5.0-bin-hadoop3/bin/sparkR.cmd
spark-3.5.0-bin-hadoop3/bin/spark-shell2.cmd
spark-3.5.0-bin-hadoop3/bin/load-spark-env.cmd
spark-3.5.0-bin-hadoop3/bin/run-example
spark-3.5.0-bin-hadoop3/bin/sparkR2.cmd
spark-3.5.0-bin-hadoop3/bin/beeline.cmd
spark-3.5.0-bin-hadoop3/bin/docker-image-tool.sh
spark-3.5.0-bin-hadoop3/bin/spark-sql.cmd
spark-3.5.0-bin-hadoop3/bin/sparkR
spark-3.5.0-bin-hadoop3/bin/spark-submit.cmd
spark-3.5.0-bin-hadoop3/bin/find-spark-home.cmd
spark-3.5.0-bin-hadoop3/bin/run-example.cmd
spark-3.5.0-bin-hadoop3/bin/spark-connect-shell
spark-3.5.0-bin-hadoop3/bin/spark-sql2.cmd
spark-3.5.0-bin-hadoop3/bin/spark-shell.cmd
spark-3.5.0-bin-hadoop3/bin/spark-class2.cmd
spark-3.5.0-bin-hadoop3/bin/spark-class
spark-3.5.0-bin-hadoop3/bin/spark-class.cmd
spark-3.5.0-bin-hadoop3/bin/spark-submit2.cmd
spark-3.5.0-bin-hadoop3/bin/spark-shell
spark-3.5.0-bin-hadoop3/bin/find-spark-home
```

Step 9: sudo mv spark-3.5.0-bin-hadoop3 /opt/spark

```
spark-3.5.0-bin-hadoop3/bin/find-spark-home
siddhesh@DESKTOP-74T5BGT:~$ sudo mv spark-3.5.0-bin-hadoop3 /opt/spark
siddhesh@DESKTOP-74T5BGT:~$
```

Step 10: Set Environment Variables

Edit .bashrc:

nano ~/.bashrc

Add at the end:

export SPARK_HOME=/opt/spark

export PATH=$PATH:$SPARK_HOME/bin

export PYSPARK_PYTHON=python3

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

```
  fi
fi
# Set Hadoop and Java paths
export HADOOP_HOME=/usr/local/hadoop
export JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-amd64

# Define Hadoop component paths
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native


# Add Hadoop to system PATH
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
# Set Hadoop JVM options
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"

export SPARK_HOME=/opt/spark
export PATH=$PATH:$SPARK_HOME/bin
export PYSPARK_PYTHON=python3
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Step 11: Apply changes:

source ~/.bashrc

```
siddhesh@DESKTOP-74T5BGT:~$ nano ~/.bashrc
siddhesh@DESKTOP-74T5BGT:~$ source ~/.bashrc
siddhesh@DESKTOP-74T5BGT:~$
```

Step 12: Verify Spark Installation

spark-shell

```
siddhesh@DESKTOP-74T5BGT:~$ source ~/.bashrc
siddhesh@DESKTOP-74T5BGT:~$ spark-shell
26/02/15 20:26:34 WARN Utils: Your hostname, DESKTOP-74T5BGT resolves to a loopback address: 127.0.1.1; using 172.23.24.114 instead (on interface eth0)
26/02/15 20:26:34 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/02/15 20:26:47 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://172.23.24.114:4040
Spark context available as 'sc' (master = local[*], app id = local-1771167408991).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.0
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.30)
Type in expressions to have them evaluated.
Type :help for more information.
```

Step 13: **Start PySpark on WSL**

pyspark

You should see:

SparkSession available as ‘spark’

```
siddhesh@DESKTOP-74T5BGT:~$ pyspark
Python 3.10.12 (main, Jan 26 2026, 14:55:28) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
26/02/15 20:28:16 WARN Utils: Your hostname, DESKTOP-74T5BGT resolves to a loopback address: 127.0.1.1; using 172.23.24.114 instead (on interface eth0)
26/02/15 20:28:16 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/02/15 20:28:20 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.0
      /_/

Using Python version 3.10.12 (main, Jan 26 2026 14:55:28)
Spark context Web UI available at http://172.23.24.114:4040
Spark context available as 'sc' (master = local[*], app id = local-1771167501816).
SparkSession available as 'spark'.
```

Step 14: **Install Required Libraries**

pip3 install pyspark matplotlib pandas

```
siddhesh@DESKTOP-74T5BGT:~$ pip3 install pyspark matplotlib pandas
Defaulting to user installation because normal site-packages is not writeable
Collecting pyspark
  Downloading pyspark-4.1.1.tar.gz (455.4 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 455.4/455.4 MB 2.4 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None)) after connection broken by 'ProtocolError('Connection aborted.', RemoteDisconnected('Remot
esponse'))': /simple/matplotlib/
Collecting matplotlib
  Downloading matplotlib-3.10.8-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.7 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 8.7/8.7 MB 5.3 MB/s eta 0:00:00
Collecting pandas
  Downloading pandas-2.3.3-cp310-cp310-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 2.2 MB/s eta 0:00:00
Collecting py4j<0.10.9.10,>=0.10.9.7
  Downloading py4j-0.10.9.9-py2.py3-none-any.whl (203 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 203.0/203.0 KB 853.0 kB/s eta 0:00:00
Collecting pyparsing>=3
  Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 122.8/122.8 KB 108.9 kB/s eta 0:00:00
Collecting cycler>=0.10
  Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Collecting kiwisolver>=1.3.1
  Downloading kiwisolver-1.4.9-cp310-cp310-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (1.6 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.6/1.6 MB 5.2 MB/s eta 0:00:00
Collecting packaging>=20.0
  Downloading packaging-26.0-py3-none-any.whl (74 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 74.4/74.4 KB 5.2 MB/s eta 0:00:00
Collecting contourpy>=1.0.1
  Downloading contourpy-1.3.2-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (325 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 325.0/325.0 KB 4.8 MB/s eta 0:00:00
Collecting numpy>=1.23
  Downloading numpy-2.2.6-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (16.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 16.8/16.8 MB 1.6 MB/s eta 0:00:00
```

Step 15: **Create python file**

nano demo.py

```
from pyspark.sql import SparkSession

import matplotlib.pyplot as plt


spark = SparkSession.builder.appName("StudentMarks").getOrCreate()


data = [
```

```
    ("A", 85),
    ("B", 78),
    ("C", 90),
    ("D", 60),
    ("E", 72)
]

df = spark.createDataFrame(data, ["Student", "Marks"])
pdf = df.toPandas()

plt.bar(pdf["Student"], pdf["Marks"])
plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.savefig("/mnt/c/Users/Public/marks.png")
print("Graph saved as marks.png")

spark.stop()
```

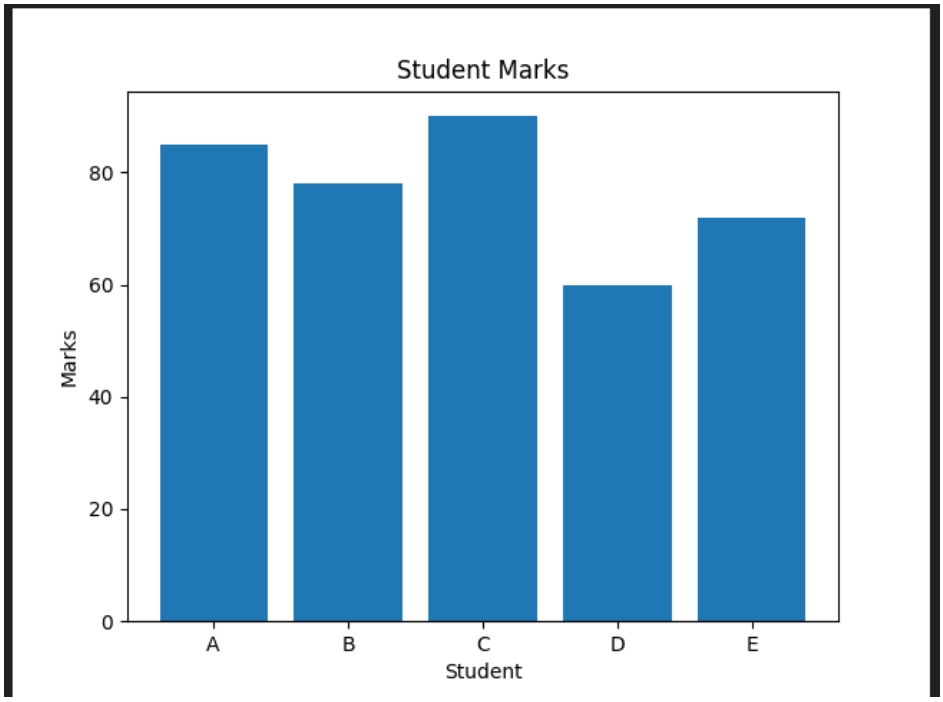
siddhesh@DESKTOP-74T5BGT: ~

```
  GNU nano 6.2
from pyspark.sql import SparkSession
import matplotlib.pyplot as plt

spark = SparkSession.builder.appName("StudentMarks").getOrCreate()

data = [
    ("A", 85),
    ("B", 78),
    ("C", 90),
    ("D", 60),
    ("E", 72)
]

df = spark.createDataFrame(data, ["Student", "Marks"])
pdf = df.toPandas()

plt.bar(pdf["Student"], pdf["Marks"])
plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.savefig("/mnt/c/Users/Public/marks.png")
print("Graph saved as marks.png")

spark.stop()
```

Step 16: Execute: spark-submit demo.py

```
siddhesh@DESKTOP-74T5BGT:~$ spark-submit demo.py
26/02/15 20:35:06 WARN Utils: Your hostname, DESKTOP-74T5BGT resolves to a loopback address: 127.0.1.1; using 172.23.24.114 instead (on interface eth0)
26/02/15 20:35:06 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
26/02/15 20:35:11 INFO SparkContext: Running Spark version 3.5.0
26/02/15 20:35:11 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
26/02/15 20:35:11 INFO SparkContext: Java version 11.0.30
26/02/15 20:35:11 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/02/15 20:35:11 INFO ResourceUtils: ==============================================================
26/02/15 20:35:11 INFO ResourceUtils: No custom resources configured for spark.driver.
26/02/15 20:35:11 INFO ResourceUtils: ==============================================================
26/02/15 20:35:11 INFO SparkContext: Submitted application: StudentMarks
26/02/15 20:35:11 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script:
Heap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/02/15 20:35:11 INFO ResourceProfile: Limiting resource is cpu
26/02/15 20:35:11 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/02/15 20:35:12 INFO SecurityManager: Changing view acls to: siddhesh
26/02/15 20:35:12 INFO SecurityManager: Changing modify acls to: siddhesh
26/02/15 20:35:12 INFO SecurityManager: Changing view acls groups to:
26/02/15 20:35:12 INFO SecurityManager: Changing modify acls groups to:
```

Step 17: Display Graph on Windows from WSL



Practical No. 5

Aim: Write a Spark code for the given application and handle error and recovery of data.

Step 1: nano student_marks.py

```
from pyspark.sql import SparkSession


# ------------------------
# Create Spark Session
# ------------------------
try:
    spark = SparkSession.builder \
        .appName("StudentMarksApp") \
        .getOrCreate()
    print("Spark session created")
except:
    print("Error creating Spark session")
    exit()


# ------------------------
# Create Data
# ------------------------
try:
    data = [
        ("A", 85),
        ("B", 45),
        ("C", 90),
        ("D", 60)
    ]

    df = spark.createDataFrame(data, ["Student", "Marks"])
    df.show()
```

```
except:
    print("Error creating DataFrame")
    spark.stop()
    exit()

# ------------------------
# Process Data
# ------------------------
try:
    passed_df = df.filter(df.Marks >= 50)
    passed_df.show()
except:
    print("Error processing data")

# ------------------------
# Save Data (Recovery)
# ------------------------
try:
    passed_df.write.mode("overwrite").csv("passed_students")
    print("Data saved successfully")
except:
    print("Error saving data")

# ------------------------
# Recover Data
# ------------------------
try:
    recovered_df = spark.read.csv(
        "passed_students", header=False, inferSchema=True
    ).toDF("Student", "Marks")
```

```
    print("Recovered Data:")
    recovered_df.show()
except:
    print("Error recovering data")


# ------------------------
# Stop Spark
# ------------------------
spark.stop()
print("Spark stopped")
```

```
siddhesh@DESKTOP-74T5BGT: ~
  GNU nano 6.2                                                    student_marks.py *
        ("D", 60)
    ]

    df = spark.createDataFrame(data, ["Student", "Marks"])
    df.show()
except:
    print("Error creating DataFrame")
    spark.stop()
    exit()

# ------------------------
# Process Data
# ------------------------
try:
    passed_df = df.filter(df.Marks >= 50)
    passed_df.show()
except:
    print("Error processing data")

# ------------------------
# Save Data (Recovery)
# ------------------------
try:
    passed_df.write.mode("overwrite").csv("passed_students")
    print("Data saved successfully")
except:
    print("Error saving data")

# ------------------------
# Recover Data
# ------------------------
try:
    recovered_df = spark.read.csv(
        "passed_students", header=False, inferSchema=True
    ).toDF("Student", "Marks")

    print("Recovered Data:")
    recovered_df.show()
except:
    print("Error recovering data")

# ------------------------
# Stop Spark
# ------------------------
spark.stop()
print("Spark stopped")
```

Step 2: spark-submit student_marks.py

```
siddhesh@DESKTOP-74T5BGT:~$ spark-submit students_marks.py
26/02/15 20:47:06 WARN Utils: Your hostname, DESKTOP-74T5BGT resolves to a loopback address: 127.0.1.1; using 172.23.24.114 instead (on interface eth0)
26/02/15 20:47:06 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
python3: can't open file '/home/siddhesh/students_marks.py': [Errno 2] No such file or directory
26/02/15 20:47:10 INFO ShutdownHookManager: Shutdown hook called
26/02/15 20:47:10 INFO ShutdownHookManager: Deleting directory /tmp/spark-bfab2f5f-27fc-44c1-8f37-659a6216b741
siddhesh@DESKTOP-74T5BGT:~$ spark-submit student_marks.py
26/02/15 20:48:04 WARN Utils: Your hostname, DESKTOP-74T5BGT resolves to a loopback address: 127.0.1.1; using 172.23.24.114 instead (on interface eth0)
26/02/15 20:48:04 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
26/02/15 20:48:08 INFO SparkContext: Running Spark version 3.5.0
26/02/15 20:48:08 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
26/02/15 20:48:08 INFO SparkContext: Java version 11.0.30
26/02/15 20:48:08 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/02/15 20:48:09 INFO ResourceUtils: ==============================================================
26/02/15 20:48:09 INFO ResourceUtils: No custom resources configured for spark.driver.
26/02/15 20:48:09 INFO ResourceUtils: ==============================================================
```

```
Spark session created
26/02/15 20:48:15 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/02/15 20:48:15 INFO SharedState: Warehouse path is 'file:/home/siddhesh/spark-warehouse'.
26/02/15 20:48:25 INFO CodeGenerator: Code generated in 788.771706 ms
26/02/15 20:48:25 INFO SparkContext: Starting job: showString at NativeMethodAccessorImpl.java:0
26/02/15 20:48:25 INFO DAGScheduler: Got job 0 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/02/15 20:48:25 INFO DAGScheduler: Final stage: ResultStage 0 (showString at NativeMethodAccessorImpl.java:0)
26/02/15 20:48:25 INFO DAGScheduler: Parents of final stage: List()
26/02/15 20:48:25 INFO DAGScheduler: Missing parents: List()
26/02/15 20:48:25 INFO DAGScheduler: Submitting ResultStage 0 (MapPartitionsRDD[6] at showString at NativeMethodAccessorImpl.java:0), which has no missing parents
26/02/15 20:48:26 INFO MemoryStore: Block broadcast_0 stored as values in memory (estimated size 12.6 KiB, free 434.4 MiB)
26/02/15 20:48:26 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 6.6 KiB, free 434.4 MiB)
26/02/15 20:48:26 INFO BlockManagerInfo: Added broadcast_0_piece0 in memory on 172.23.24.114:46663 (size: 6.6 KiB, free: 434.4 MiB)
26/02/15 20:48:26 INFO SparkContext: Created broadcast 0 from broadcast at DAGScheduler.scala:1580
```

```
26/02/15 20:48:41 INFO FileScanRDD: Reading File path: file:///home/siddhesh/passed_students/part-00002-42f79623-86de-4053-a651-d5a93043c32e-c000.csv, range: 0-5, partition
26/02/15 20:48:41 INFO BlockManagerInfo: Removed broadcast_8_piece0 on 172.23.24.114:46663 in memory (size: 12.6 KiB, free: 434.4 MiB)
26/02/15 20:48:42 INFO Executor: Finished task 0.0 in stage 8.0 (TID 17). 1533 bytes result sent to driver
26/02/15 20:48:42 INFO TaskSetManager: Finished task 0.0 in stage 8.0 (TID 17) in 68 ms on 172.23.24.114 (executor driver) (1/2)
26/02/15 20:48:42 INFO Executor: Finished task 1.0 in stage 8.0 (TID 18). 1533 bytes result sent to driver
26/02/15 20:48:42 INFO TaskSetManager: Finished task 1.0 in stage 8.0 (TID 18) in 112 ms on 172.23.24.114 (executor driver) (2/2)
26/02/15 20:48:42 INFO TaskSchedulerImpl: Removed TaskSet 8.0, whose tasks have all completed, from pool
26/02/15 20:48:42 INFO DAGScheduler: ResultStage 8 (showString at NativeMethodAccessorImpl.java:0) finished in 0.208 s
26/02/15 20:48:42 INFO DAGScheduler: Job 8 is finished. Cancelling potential speculative or zombie tasks for this job
26/02/15 20:48:42 INFO TaskSchedulerImpl: Killing all running tasks in stage 8: Stage finished
26/02/15 20:48:42 INFO DAGScheduler: Job 8 finished: showString at NativeMethodAccessorImpl.java:0, took 0.257406 s
26/02/15 20:48:42 INFO SparkContext: SparkContext is stopping with exitCode 0.
+-------+-----+
|Student|Marks|
+-------+-----+
|      D|   60|
|      A|   85|
|      C|   90|
+-------+-----+

26/02/15 20:48:42 INFO SparkUI: Stopped Spark web UI at http://172.23.24.114:4040
26/02/15 20:48:42 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/02/15 20:48:42 INFO MemoryStore: MemoryStore cleared
```

```
26/02/15 20:48:42 INFO SparkUI: Stopped Spark web UI at http://172.23.24.114:4040
26/02/15 20:48:42 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/02/15 20:48:42 INFO MemoryStore: MemoryStore cleared
26/02/15 20:48:42 INFO BlockManager: BlockManager stopped
26/02/15 20:48:42 INFO BlockManagerMaster: BlockManagerMaster stopped
26/02/15 20:48:42 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/02/15 20:48:42 INFO SparkContext: Successfully stopped SparkContext
Spark stopped
26/02/15 20:48:43 INFO ShutdownHookManager: Shutdown hook called
26/02/15 20:48:43 INFO ShutdownHookManager: Deleting directory /tmp/spark-3d7e3d46-d818-49b3-9f8d-4f90123d2b35
26/02/15 20:48:43 INFO ShutdownHookManager: Deleting directory /tmp/spark-0882e8d1-ab68-4efb-a48c-eb05882e9827/pyspark-7f8db7c7-4010-4e25-ac56-56e7008aacf2
26/02/15 20:48:43 INFO ShutdownHookManager: Deleting directory /tmp/spark-0882e8d1-ab68-4efb-a48c-eb05882e9827
siddhesh@DESKTOP-74T5BGT:~$
```

Practical No. 6

Aim: Write a pig script for solving counting problem

Step 1: Java –version

```
siddhesh@DESKTOP-74T5BGT:~$ java --version
openjdk 11.0.30 2026-01-20
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu122.04, mixed mode, sharing)
siddhesh@DESKTOP-74T5BGT:~$
```

Step 2: Hadoop version

```
siddhesh@DESKTOP-74T5BGT:~$ hadoop version
Hadoop 3.3.1
Source code repository https://github.com/apache/hadoop.git -r a3b9c37a397ad4188041dd80621bdeefc46885f2
Compiled by ubuntu on 2021-06-15T05:13Z
Compiled with protoc 3.7.1
From source with checksum 88a4ddb2299aca054416d6b7f81ca55
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.1.jar
siddhesh@DESKTOP-74T5BGT:~$
```

Step 3: Start Hadoop

Command 1: su siddhesh

Command 2: sudo service ssh restart

Command 3: ssh localhost

Command 4: start-all.sh

```
siddhesh@DESKTOP-74T5BGT:~$ su siddhesh
Password:
siddhesh@DESKTOP-74T5BGT:~$ sudo service ssh restart
[sudo] password for siddhesh:
siddhesh@DESKTOP-74T5BGT:~$ ssh localhost
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Feb 15 20:56:20 IST 2026

  System load:  0.02               Processes:             52
  Usage of /:   0.9% of 1006.85GB  Users logged in:       1
  Memory usage: 27%                IPv4 address for eth0: 172.23.24.114
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Feb 15 20:12:12 IST 2026

  System load:  1.4                Processes:             54
  Usage of /:   0.6% of 1006.85GB  Users logged in:       1
  Memory usage: 27%                IPv4 address for eth0: 172.23.24.114
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Sun Feb 15 19:40:10 2026 from 127.0.0.1
siddhesh@DESKTOP-74T5BGT:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as siddhesh in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
```

Step 4: Change the current working directory to / home/siddhesh

Command 5: cd /home/siddhesh

```
localhost: nodemanager is running as process 5493.  Stop it first and ensure /tmp/hadoop-
siddhesh@DESKTOP-74T5BGT:~$ cd /home/siddhesh
siddhesh@DESKTOP-74T5BGT:~$ cd home/siddhesh
```

Step 5: Create a text file sample.txt:

Command 6: sudo nano sample.txt

1,Rasika,20000,Lecturer

2,Hrishi,25000,Accountant

3,Rahul,30000,SoftwareEngineer

```
siddhesh@DESKTOP-74T5BGT: ~
  GNU nano 6.2
1,Rasika,20000,Lecturer
2,Hrishi,25000,Accountant
3,Rahul,30000,SoftwareEngineer
```

Step 6: Change the current working directory to /usr/local

cd /usr/local

```
[sudo] password for siddhesh:
siddhesh@DESKTOP-74T5BGT:~$ cd /usr/local
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 7: Download the new release of Apache Pig.

In my case I have downloaded the pig-0.17.0.tar.gz version of Pig which is latest and about 220MB in size

Command: sudo wget https://downloads.apache.org/pig/pig-0.17.0/pig-0.17.0.tar.gz

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo wget https://downloads.apache.org/pig/pig-0.17.0/pig-0.17.0.tar.gz
--2026-02-15 21:05:53--  https://downloads.apache.org/pig/pig-0.17.0/pig-0.17.0.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.214.104, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 230606579 (220M) [application/x-gzip]
Saving to: 'pig-0.17.0.tar.gz'

pig-0.17.0.tar.gz                   100%[===================================================================================================>] 219.92M   375KB/s    in 6m 12s

2026-02-15 21:12:07 (605 KB/s) - 'pig-0.17.0.tar.gz' saved [230606579/230606579]
```

Step 8: Once the download process is complete, untar the compressed pig package:

Command: sudo tar -xvzf pig-0.17.0.tar.gz

```
pig-0.17.0/test/resources/org/apache/pig/test/ClassLoaderSanityCheck.java
pig-0.17.0/test/resources/org/apache/pig/test/RegisteredJarVisibilityLoader.java
pig-0.17.0/test/resources/org/apache/pig/test/RegisteredJarVisibilitySchema.java
pig-0.17.0/test/smoke-tests
pig-0.17.0/test/unit-tests
pig-0.17.0/tutorial/build.xml
pig-0.17.0/tutorial/data/excite-small.log
pig-0.17.0/tutorial/data/excite.log.bz2
pig-0.17.0/tutorial/scripts/script1-hadoop.pig
pig-0.17.0/tutorial/scripts/script1-local.pig
pig-0.17.0/tutorial/scripts/script2-hadoop.pig
pig-0.17.0/tutorial/scripts/script2-local.pig
pig-0.17.0/tutorial/src/org/apache/pig/tutorial/ExtractHour.java
pig-0.17.0/tutorial/src/org/apache/pig/tutorial/NGramGenerator.java
pig-0.17.0/tutorial/src/org/apache/pig/tutorial/NonURLDetector.java
pig-0.17.0/tutorial/src/org/apache/pig/tutorial/ScoreGenerator.java
pig-0.17.0/tutorial/src/org/apache/pig/tutorial/ToLower.java
pig-0.17.0/tutorial/src/org/apache/pig/tutorial/TutorialTest.java
pig-0.17.0/tutorial/src/org/apache/pig/tutorial/TutorialUtil.java
pig-0.17.0/bin/pig
pig-0.17.0/bin/pig.cmd
pig-0.17.0/bin/pig.py
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 9: Rename the extracted file from pig-0.17.0 to pig

Command: sudo mv pig-0.17.0 pig

```
pig-0.17.0/bin/pig.cmd
pig-0.17.0/bin/pig.py
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo mv pig-0.17.0 pig
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 10: Define or change permissions or modes on files and limit access to only those who are allowed access.

Checking the permission of pig folder

Command: ls -l

Command: sudo chmod 777 pig

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ ls -l
total 816252
drwxr-xr-x  2 root     root        4096 Nov 28 17:09 bin
drwxr-xr-x  2 root     root        4096 Jan  7  2025 etc
drwxr-xr-x  2 root     root        4096 Jan  7  2025 games
drwxr-xr-x 11 siddhesh hadoop      4096 Feb 15 19:34 hadoop
-rw-r--r--  1 root     root   605187279 Jun 15  2021 hadoop-3.3.1.tar.gz
drwxr-xr-x  2 root     root        4096 Jan  7  2025 include
drwxr-xr-x  4 root     root        4096 Aug  8  2025 lib
lrwxrwxrwx  1 root     root           9 Jan  7  2025 man -> share/man
drwxr-xr-x 16 root     root        4096 Jun  2  2017 pig
-rw-r--r--  1 root     root   230606579 Jun 16  2017 pig-0.17.0.tar.gz
drwxr-xr-x  2 root     root        4096 Jan  7  2025 sbin
drwxr-xr-x  5 root     root        4096 Jan  7  2025 share
drwxr-xr-x  2 root     root        4096 Jan  7  2025 src
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo chmod 777 pig
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 11: Change the current working directory to / home/siddhesh

Command: cd /home/siddhesh

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo chmod 777 pig
siddhesh@DESKTOP-74T5BGT:/usr/local$ cd /home/siddhesh
siddhesh@DESKTOP-74T5BGT:~$
```

Step 12: The $HIVE_HOME environment variable needs to direct the client shell to the apache-pig directory. Edit the .bashrc shell configuration file using a text editor of your choice (we will be using nano):

Command: nano .bashrc

Append the following Pig environment variables to the .bashrc file:

export PATH=$PATH:$HADOOP_HOME/bin

export PATH=$PATH:$HADOOP_HOME/sbin

export PATH=$PATH:/usr/local/pig/bin

export PIG_HOME=/usr/local/pig

export PIG_CLASSPATH=/usr/local/hadoop/etc/hadoop

The Hadoop environment variables are located within the same file.

```
siddhesh@DESKTOP-74T5BGT: ~
  GNU nano 6.2                                                    .bashrc *
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
# Set Hadoop and Java paths
export HADOOP_HOME=/usr/local/hadoop
export JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-amd64

# Define Hadoop component paths
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native


# Add Hadoop to system PATH
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
# Set Hadoop JVM options
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"

export SPARK_HOME=/opt/spark
export PATH=$PATH:$SPARK_HOME/bin
export PYSPARK_PYTHON=python3
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export PATH=$PATH:/usr/local/pig/bin
export PIG_HOME=/usr/local/pig
export PIG_CLASSPATH=/usr/local/hadoop/etc/hadoop
```

Step 13: Save and exit the .bashrc file once you add the Pig variables. Apply the changes to the current environment with the following command:

Command: source .bashrc

```
siddhesh@DESKTOP-74T5BGT:~$ source .bashrc
siddhesh@DESKTOP-74T5BGT:~$
```

Step 14: We have successfully installed Apache Pig in Ubuntu. Now check the Pig version to verify the installation. Execute the below command:

Command: pig -version

```
siddhesh@DESKTOP-74T5BGT:~$ pig --version
Apache Pig version 0.17.0 (r1797386)
compiled Jun 02 2017, 15:41:58
```

Step 15: Start Apache Pig.

When we start Apache Pig, it opens a grunt shell.

We can start Pig in one of the following two modes mentioned below:

• Local Mode

• Cluster Mode

To start using pig in local mode '-x local' option is used whereas while executing only "pig" command without any options, Pig starts in the cluster mode.

While running pig in local mode, it can only access files present on the local file system.

Whereas, on starting pig in cluster mode pig can access files present in HDFS.

To start Pig in Local Mode, execute the below command:

Command: pig -x local

And if you get the below output that means Pig started successfully in Local mode.

```
siddhesh@DESKTOP-74T5BGT:~$ pig -x local
2026-02-15 21:19:34,734 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-02-15 21:19:34,870 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-02-15 21:19:34,870 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-02-15 21:19:35,001 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-02-15 21:19:35,002 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/siddhesh/pig_1771170574994.log
2026-02-15 21:19:35,049 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/siddhesh/.pigbootup not found
2026-02-15 21:19:35,218 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-02-15 21:19:35,222 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-02-15 21:19:35,503 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
2026-02-15 21:19:35,543 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-default-c4d3c116-43aa-48e4-8de6-7177bd83ca34
2026-02-15 21:19:35,543 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
grunt>
```

Step 16: Load the data in the file named sample.txt as a relation named Employees.

Command:

Employees = LOAD 'sample.txt'

USING PigStorage(',')

AS (id:int, firstname:chararray, salary:int, Dept:chararray);

```
grunt> Employees = LOAD 'sample.txt'
>> USING PigStorage(',')
>> AS (id:int, firstname:chararray, salary:int, Dept:chararray);
2026-02-15 21:21:17,655 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
grunt>
```

Step 17: Display the Employees table on the screen.

The Dump operator is used to run the Pig Latin statements and display the results on the screen.

Command: Dump Employees;

```
HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.1   0.17.0  siddhesh        2026-02-15 21:22:34     2026-02-15 21:22:36     UNKNOWN

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime  AvgReduceTime
edianReducetime Alias   Feature Outputs
job_local1090709583_0001        1       0       n/a     n/a     n/a     n/a     0       0       0       0       Employees
AP_ONLY file:/tmp/temp507678488/tmp-860122748,

Input(s):
Successfully read 4 records from: "file:///home/siddhesh/sample.txt"

Output(s):
Successfully stored 4 records in: "file:/tmp/temp507678488/tmp-860122748"

Counters:
Total records written : 4
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1090709583_0001


2026-02-15 21:22:36,450 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-02-15 21:22:36,457 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-02-15 21:22:36,460 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-02-15 21:22:36,468 [main] WARN  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning ACCESSING_NON_EXISTENT_FIELD 3 time(s).
2026-02-15 21:22:36,469 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-02-15 21:22:36,473 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
2026-02-15 21:22:36,474 [main] WARN  org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-02-15 21:22:36,498 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-02-15 21:22:36,498 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(1,Rasika,20000,Lecturer)
(2,Hrishi,25000,Accountant)
(3,Rahul,30000,SoftwareEngineer)
(,,,)
```

Step 18: Arrange the tuples of the relation in descending order, based on salary and store it as Employees_order.

Command: Employees_order = ORDER Employees BY salary DESC;

```
(2,Hrishi,25000,Accountant)
(3,Rahul,30000,SoftwareEngineer)
(,,,)
grunt> Employees_order = ORDER Employees BY salary DESC;
grunt>
```

Step 19: Command: quit

```
(,,,)
grunt> quit
2026-02-15 21:24:24,537 [main] INFO  org.apache.pig.Main - Pig script completed in 4 minutes, 50 seconds and 148 milliseconds (290148 ms)
siddhesh@DESKTOP-74T5BGT:~$
```

Step 20: Running in HDFS mode (pig can access data on HDFS).

Move sample.txt to HDFS.

Command: hdfs dfs -put sample.txt /

```
2026-02-15 21:24:24,537 [main] INFO  org.apache.pig.Main - Pig script completed in 4 minutes, 50 seconds and 148 milliseconds (290148 ms)
siddhesh@DESKTOP-74T5BGT:~$ hdfs dfs -put sample.txt /
2026-02-15 21:25:10,377 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
siddhesh@DESKTOP-74T5BGT:~$
```

Step 21: Check sample.txt got saved to HDFS or not by listing files of HDFS.

Command: hdfs dfs -ls /

```
siddhesh@DESKTOP-74T5BGT:~$ hdfs dfs -ls /
2026-02-15 21:26:02,244 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r--   1 siddhesh supergroup         82 2026-02-15 21:25 /sample.txt
siddhesh@DESKTOP-74T5BGT:~$
```

Step 22: Start pig using pig command.

Command: pig

```
siddhesh@DESKTOP-74T5BGT:~$ pig
2026-02-15 21:26:37,681 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-02-15 21:26:37,684 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-02-15 21:26:37,689 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-02-15 21:26:37,791 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-02-15 21:26:37,791 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/siddhesh/pig_1771170997780.log
2026-02-15 21:26:37,818 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/siddhesh/.pigbootup not found
2026-02-15 21:26:38,346 [main] WARN  org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-02-15 21:26:38,377 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-02-15 21:26:38,377 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://127.0.0.1:54310
2026-02-15 21:26:39,347 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-default-d3c1582d-3a5e-4e6f-b4ec-278247e49607
2026-02-15 21:26:39,347 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
grunt>
```

Step 23: Load the data in the file named sample.txt as a relation named Employee.

Command 25:

Emp = LOAD ‘hdfs://localhost:9000/sample.txt’ USING PigStorage(‘,’);

(NOTE: Give a space after and before assignment operator “=”)

```
grunt> Emp = LOAD 'hdfs://localhost:9000/sample.txt' USING PigStorage(',');
grunt>
```

Step 24: Display the Employee table on the screen.

Command: Dump Emp;

```
        ... 46 more
        hdfs://127.0.0.1:54310/tmp/temp741675343/tmp1183971130,

Input(s):
Failed to read data from "hdfs://localhost:9000/sample.txt"

Output(s):
Failed to produce result in "hdfs://127.0.0.1:54310/tmp/temp741675343/tmp1183971130"

Counters:
Total records written : 0
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1809271564_0001


2026-02-15 21:28:44,062 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Failed!
2026-02-15 21:28:44,079 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1066: Unable to open iterator for alias Emp. Backend error : java.lang.IllegalStateExceptio
Details at logfile: /home/siddhesh/pig_1771170997780.log
grunt>
```

Practical No. 7

Aim: Install Perform HBase and use the HBase Data model Store and retrieve data.

Step 1: Start the second user

su siddhesh

```
siddhesh@DESKTOP-74T5BGT:~$ su siddhesh
Password:
siddhesh@DESKTOP-74T5BGT:~$
```

Step 2: Move directory to usr/local

cd /usr/local

```
siddhesh@DESKTOP-74T5BGT:~$ cd /usr/local
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 3: Download the file hbase

sudo wget https://archive.apache.org/dist/hbase/2.5.10/hbase-2.5.10-bin.tar.gz

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo wget https://archive.apache.org/dist/hbase/2.5.10/hbase-2.5.10-bin.tar.gz
[sudo] password for siddhesh:
--2026-02-15 21:35:27--  https://archive.apache.org/dist/hbase/2.5.10/hbase-2.5.10-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 322731236 (308M) [application/x-gzip]
Saving to: ‘hbase-2.5.10-bin.tar.gz’

hbase-2.5.10-bin.tar.gz             100%[===================================================================>] 307.78M   470KB/s    in 10m 19s

2026-02-15 21:45:49 (509 KB/s) - ‘hbase-2.5.10-bin.tar.gz’ saved [322731236/322731236]

siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 4: Unzip the file

sudo tar -xvzf hbase-2.5.10-bin.tar.gz

```
hbase-2.5.10/lib/jdk11/FastInfoset-1.2.16.jar
hbase-2.5.10/lib/jdk11/ha-api-3.1.12.jar
hbase-2.5.10/lib/jdk11/saaj-impl-1.5.1.jar
hbase-2.5.10/lib/jdk11/jaxws-tools-2.3.2.jar
hbase-2.5.10/lib/jdk11/jaxb-xjc-2.3.2.jar
hbase-2.5.10/lib/jdk11/jaxb-jxc-2.3.2.jar
hbase-2.5.10/lib/jdk11/jaxws-eclipselink-plugin-2.3.2.jar
hbase-2.5.10/lib/jdk11/jakarta.mail-api-1.6.3.jar
hbase-2.5.10/lib/jdk11/jakarta.persistence-api-2.2.2.jar
hbase-2.5.10/lib/jdk11/org.eclipse.persistence.moxy-2.7.4.jar
hbase-2.5.10/lib/jdk11/org.eclipse.persistence.core-2.7.4.jar
hbase-2.5.10/lib/jdk11/org.eclipse.persistence.asm-2.7.4.jar
hbase-2.5.10/lib/jdk11/sdo-eclipselink-plugin-2.3.2.jar
hbase-2.5.10/lib/jdk11/org.eclipse.persistence.sdo-2.7.4.jar
hbase-2.5.10/lib/jdk11/commonj.sdo-2.1.1.jar
hbase-2.5.10/lib/jdk11/release-documentation-2.3.2-docbook.zip
hbase-2.5.10/lib/jdk11/samples-2.3.2.zip
hbase-2.5.10/lib/jdk11/jakarta.xml.ws-api-2.3.2.jar
hbase-2.5.10/lib/jdk11/jakarta.xml.bind-api-2.3.2.jar
hbase-2.5.10/lib/jdk11/jakarta.xml.soap-api-1.4.1.jar
hbase-2.5.10/lib/jdk11/jakarta.jws-api-1.1.1.jar
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 5: Change file name to hbase

sudo mv hbase-2.5.10 hbase

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ sudo mv hbase-2.5.10 hbase
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 6: Open the bashrc file and add the content

nano ~/.bashrc

Add this content in it

export HBASE_HOME=/usr/local/hbase

export PATH=$PATH:$HBASE_HOME/bin

```
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export PATH=$PATH:/usr/local/pig/bin
export PIG_HOME=/usr/local/pig
export PIG_CLASSPATH=/usr/local/hadoop/etc/hadoop

export HBASE_HOME=/usr/local/hbase
export PATH=$PATH:$HBASE_HOME/bin

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E
```

Step 7: Run the bashrc file

source ~/.bashrc

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ nano ~/.bashrc
siddhesh@DESKTOP-74T5BGT:/usr/local$ source ~/.bashrc
siddhesh@DESKTOP-74T5BGT:/usr/local$
```

Step 8: Change directory to hbase conf folder

cd /usr/local/hbase/conf

```
siddhesh@DESKTOP-74T5BGT:/usr/local$ cd /usr/local/hbase/conf
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/conf$
```

Step 9: Edit the hbase-site file

sudo nano hbase-site.xml

<configuration>

<property>

<name>hbase.rootdir</name>

<value>file:///usr/local/hbase/data</value>

</property>

<property>

<name>hbase.zookeeper.property.dataDir</name>

<value>/usr/local/hbase/zookeeper</value>

</property>

</configuration>

```
siddhesh@DESKTOP-74T5BGT: /usr/local/hbase/conf
  GNU nano 6.2                                          hbase-site.xml *
    this HBase installation directory.

    Running against the `LocalFileSystem`, as opposed to a distributed
    filesystem, runs the risk of data integrity issues and data loss. Normally
    HBase will refuse to run in such an environment. Setting
    `hbase.unsafe.stream.capability.enforce` to `false` overrides this behavior,
    permitting operation. This configuration is for the developer workstation
    only and __should not be used in production!__

    See also https://hbase.apache.org/book.html#standalone_dist
  -->
  <property>
    <name>hbase.cluster.distributed</name>
    <value>false</value>
  </property>
  <property>
    <name>hbase.tmp.dir</name>
    <value>./tmp</value>
  </property>
  <property>
    <name>hbase.unsafe.stream.capability.enforce</name>
    <value>false</value>
  </property>
<property>
<name>hbase.rootdir</name>
<value>file:///usr/local/hbase/data</value>
</property>
<property>
<name>hbase.zookeeper.property.dataDir</name>
<value>/usr/local/hbase/zookeeper</value>
</property>

</configuration>
```

Step 10: Edit the hbase-env file

sudo nano hbase-env.sh

```
#export HBASE_SHELL_OPTS="${HBASE_SHELL_OPTS} -Dotel.resource.attributes=service.name=hbase-shell"
#export HBASE_THRIFT_OPTS="${HBASE_THRIFT_OPTS} -Dotel.resource.attributes=service.name=hbase-thrift"
#export HBASE_ZOOKEEPER_OPTS="${HBASE_ZOOKEEPER_OPTS} -Dotel.resource.attributes=service.name=hbase-zookeeper"

#
# JDK11+ JShell
#
# Additional arguments passed to jshell invocation
# export HBASE_JSHELL_ARGS="--startup DEFAULT --startup PRINTING --startup hbase_startup.jsh"
export JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-amd64
export HBASE_CLASSPATH=${HBASE_HOME}/lib/*

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Br
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^Q Where
```

Step 11: Move directory to bin folder

cd /usr/local/hbase/bin

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/conf$ sudo nano hbase-env.sh
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/conf$ cd /usr/local/hbase/bin
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/bin$ A
```

Step 12: Type command

sudo ./start-hbase.sh

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/conf$ cd /usr/local/hbase/bin
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/bin$ sudo ./start-hbase.sh
running master, logging to /usr/local/hbase/bin/../logs/hbase-root-master-DESKTOP-74T5BGT.out
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/bin$
```

Step 13: Type command

sudo ./hbase shell

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/bin$ sudo ./hbase shell
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.5.10, ra3af60980c61fb4be31e0dcd89880f304d01098a, Thu Jul 18 22:45:17 PDT 2024
Took 0.0032 seconds
hbase:001:0>
```

Step 14: Create table

create 'users', 'info'

```
siddhesh@DESKTOP-74T5BGT:/usr/local/hbase/bin$ sudo ./hbase shell
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.5.10, ra3af60980c61fb4be31e0dcd89880f304d01098a, Thu Jul 18 22:45:17 P
Took 0.0032 seconds
hbase:001:0> create 'users', 'info'
Created table users
Took 1.4780 seconds
=> Hbase::Table - users
hbase:002:0>
```

Step 15: Add value in the table

put 'users', '1', 'info:name', 'Alice'

put 'users', '1', 'info:age', '25'

put 'users', '2', 'info:name', 'Bob'

put 'users', '2', 'info:age', '30'

```
Took 1.4780 seconds
=> Hbase::Table - users
hbase:002:0> put 'users', '1', 'info:name', 'Alice'
Took 0.4720 seconds
hbase:003:0> put 'users', '1', 'info:age', '25'
Took 0.0187 seconds
hbase:004:0> put 'users', '2', 'info:name', 'Bob'
Took 0.0080 seconds
hbase:005:0> put 'users', '2', 'info:age', '30'
Took 0.0068 seconds
hbase:006:0>
```

Step 16: Try to retrieve the data

get 'users', '1', 'info:name'

```
hbase:006:0> get 'users', '1', 'info:name'
COLUMN                                 CELL
 info:name                             timestamp=2026-02-15T22:17:04.564, value=Alice
1 row(s)
Took 0.1655 seconds
hbase:007:0>
```

Step 17: Scans and displays all rows in the users table.

scan 'users'

```
Took 0.1655 seconds
hbase:007:0> scan 'users'
ROW                                    COLUMN+CELL
 1                                     column=info:age, timestamp=2026-02-15T22:17:10.127, value=25
 1                                     column=info:name, timestamp=2026-02-15T22:17:04.564, value=Alice
 2                                     column=info:age, timestamp=2026-02-15T22:17:20.594, value=30
 2                                     column=info:name, timestamp=2026-02-15T22:17:15.983, value=Bob
2 row(s)
Took 0.0316 seconds
hbase:008:0>
```

Step 18: Deletes 'age' column for Row Key '1'.

delete 'users', '1', 'info:age'

```
 2                                     column=info:age, time
 2                                     column=info:name, tim
2 row(s)
Took 0.0316 seconds
hbase:008:0> delete 'users', '1', 'info:age'
Took 0.0122 seconds
hbase:009:0>
```

Step 19: Deletes Row Key '1' entirely.

deleteall 'users', '1'

```
hbase:008:0> delete 'users', '1', 'info:age'
Took 0.0122 seconds
hbase:009:0> deleteall 'users', '1'
Took 0.0064 seconds
hbase:010:0>
```

Step 20: Disables and then deletes the 'users' table.

disable 'users'

drop 'users'

```
Took 0.0064 seconds
hbase:010:0> disable 'users'
Took 1.2684 seconds
hbase:011:0> drop 'users'
Took 0.4092 seconds
hbase:012:0>
```

Practical No. 8

Aim: Write a Spark code to Handle the Streaming of data.

Step 1: Verify Hadoop & Spark Installation

```
siddhesh@DESKTOP-74T5BGT:~$ hadoop version
Hadoop 3.3.1
Source code repository https://github.com/apache/hadoop.git -r a3b9c37a397ad4188041dd80621bdeefc46885f2
Compiled by ubuntu on 2021-06-15T05:13Z
Compiled with protoc 3.7.1
From source with checksum 88a4ddb2299aca054416d6b7f81ca55
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.1.jar
siddhesh@DESKTOP-74T5BGT:~$ spark-submit --version
26/02/17 11:16:08 WARN Utils: Your hostname, DESKTOP-74T5BGT resolves to a loopback address: 127.0.1.1; using 172.23.24.
114 instead (on interface eth0)
26/02/17 11:16:08 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.0
      /_/

Using Scala version 2.12.18, OpenJDK 64-Bit Server VM, 11.0.30
Branch HEAD
Compiled by user ubuntu on 2023-09-09T01:53:20Z
Revision ce5ddad990373636e94071e7cef2f31021add07b
Url https://github.com/apache/spark
Type --help for more information.
siddhesh@DESKTOP-74T5BGT:~$
```

Step 2: Write a simple Spark Streaming program to count the number of words in text data received from a data server listening on a TCP socket.

```
Type --help for more information.
siddhesh@DESKTOP-74T5BGT:~$ nano streamingcode.py
```

```
siddhesh@DESKTOP-74T5BGT: ~
  GNU nano 6.2                                    streamingcode.py *
from pyspark import SparkContext
from pyspark.streaming import StreamingContext

# Initialize Spark Context
sc = SparkContext(appName="StreamingExample")

# Create Streaming Context with 5-second batch interval
ssc = StreamingContext(sc, 5)

# Define Input Stream (Example: Read from a TCP socket)
stream = ssc.socketTextStream("localhost", 9999)

# Process Data (Word Count Example)
words = stream.flatMap(lambda line: line.split(" "))
word_counts = words.map(lambda word: (word, 1)).reduceByKey(lambda a, b: a + b)

# Print Output
word_counts.pprint()

# Start Streaming
ssc.start()
ssc.awaitTermination()
```

Step 3: Start a Netcat Server (Simulates a Stream)

Command - nc -lk 9999

(Type some text and press Enter)

```
siddhesh@DESKTOP-74T5BGT:~$ nc -lk 9999
Hello spark siddhesh this side
```

Step 4: Run the PySpark Streaming Script

spark-submit ~/StreamingCode.py

Check Output in Console It should show word counts in real-time.

```
siddhesh@DESKTOP-74T5BGT: ~
siddhesh@DESKTOP-74T5BGT:~$ nc -lk 9999
Hello Siddhesh this side from spark
hello
sid
santosh
teli
sidhhh
hello my name is siddhesh and i am in msc
```

```
siddhesh@DESKTOP-74T5BGT: ~
26/02/17 11:30:41 INFO Executor: Finished task 0.0 in stage 152.0 (TID 156). 2110 bytes result sent to driver
26/02/17 11:30:41 INFO TaskSetManager: Finished task 0.0 in stage 152.0 (TID 156) in 816 ms on 172.23.24.114 (executor d
river) (3/3)
26/02/17 11:30:41 INFO TaskSchedulerImpl: Removed TaskSet 152.0, whose tasks have all completed, from pool
26/02/17 11:30:41 INFO DAGScheduler: ResultStage 152 (runJob at PythonRDD.scala:181) finished in 0.844 s
26/02/17 11:30:41 INFO DAGScheduler: Job 76 is finished. Cancelling potential speculative or zombie tasks for this job
26/02/17 11:30:41 INFO TaskSchedulerImpl: Killing all running tasks in stage 152: Stage finished
26/02/17 11:30:41 INFO DAGScheduler: Job 76 finished: runJob at PythonRDD.scala:181, took 0.849528 s
-------------------------------------------
Time: 2026-02-17 11:30:40
-------------------------------------------
('name', 1)
('i', 1)
('in', 1)
('', 1)
('hello', 1)
('and', 1)
('msc', 1)
('is', 1)
('siddhesh', 1)
('am', 1)
...

26/02/17 11:30:41 INFO JobScheduler: Finished job streaming job 1771308040000 ms.0 from job set of time 1771308040000 ms
```